SIMATS ENGINEERING

ASSESSMENT TOOL 2 - COURSE: ARTIFICIAL INTELLIGENCE (CSA17)

Comprehensive Solution & Implementation Guidelines

Task 1: Data Preparation & Inductive Learning

(a) Inductive Learning Approach & Feature Identification:

Inductive learning extracts general medical rules from a finite dataset of historical patient records. By mapping observed diagnostic clinical indicators to historical clinical outcomes, the system generalizes rules to classify unseen patients.

- **Target Variable:** Diabetes_Status (Binary: 1 for Diabetic, 0 for Non-Diabetic).
- **Key Feature 1: Glucose Level:** Directly measures blood sugar concentration; standard clinical biomarker for diabetes diagnosis.
- **Key Feature 2: BMI (Body Mass Index):** Represents metabolic health and adiposity, strongly linked to insulin resistance.
- **Key Feature 3: Family History:** Encodes genetic predisposition, critical for establishing prior probability in risk assessment.

(b) Handling Class Imbalance:

In clinical diagnostics, non-diabetic records typically dominate. Standard classifiers tend to minimize global error by bias-favoring the majority class, creating a critical risk of missing positive diabetes diagnoses.

Recommended Strategy: SMOTE (Synthetic Minority Over-sampling Technique) + Cost-Sensitive Learning (Class Weights).

- **Justification:** Pure random under-sampling discards valuable clinical variance. SMOTE synthesizes new minority examples along line segments connecting existing k-nearest neighbors. Combining this with inverse class weighting inside loss functions forces the optimization path to heavily penalize missing true diabetic cases.

(c) Dataset Split and Preprocessing:

- **70:30 Ratio Justification:** Provides 70% of data to expose the learner to sufficient variations of high-dimensional interactions (such as interactions between age, glucose, and family history), while preserving a large 30% holdout set to ensure the calculated evaluation metrics possess tight confidence intervals and reflect authentic clinical behavior.

- **Normalization (Min-Max Scaling):** Transforms features like Age and Blood Pressure to a uniform [0, 1] range. This prevents attributes with higher absolute values from improperly dominating distance or split metrics.
- **Encoding (One-Hot / Label Encoding):** Converts categorical items like Family History into numerical indices. This maps conceptual metadata directly into algebraic dimensions compatible with math matrices.

Task 2: Decision Tree for Diagnosis

(a) Decision Tree Topology & Root Node Selection:

Using Gini Impurity, the root node selection prioritizes the feature that maximizes impurity reduction. For example, if the baseline Gini is 0.45, splitting on *Glucose > 140 mg/dL* drops the child node weighted Gini down to 0.22, yielding the highest overall Information Gain.

Three-Level Diagnostic Tree Structure:

Level 0 (Root): [Glucose ≤ 140 mg/dL]

Level 1 (Left Child): [BMI ≤ 30] → **Level 2:** [Age ≤ 45] → Class: Normal (Low Risk)

Level 1 (Right Child): [Family History = Yes] → **Level 2:** [Glucose ≤ 165] → Class: Diabetic (High Risk)

(b) Model Evaluation & False Negative Mitigation:

Metric	Value	Clinical Significance
Accuracy	86.5%	Overall structural correctness of diagnostics.
Precision	82.1%	Minimizes false alarms / unnecessary patient anxiety.
Recall (Sensitivity)	76.4%	Critical measure. Captures true diabetic individuals.
F1-Score	79.1%	Harmonic balance between Precision and Sensitivity.

Reducing False Negatives (Clinical Justification): False Negatives mean a diabetic patient is mistakenly classified as healthy, leading to delayed medical care and microvascular damage. To reduce False Negatives, we must lower the classification probability threshold from 0.5 to 0.35, or apply a heavier cost-penalty ratio (e.g., 5:1) on False Negatives during node optimization.

(c) Pruning Strategies & Deployment Choice:

• **Pre-pruning:** Halts tree development prematurely using arbitrary thresholds (e.g., max_depth=5). This is computationally efficient but vulnerable to underfitting because it misses multi-feature combinations that appear valuable only after consecutive splits.

• **Post-pruning (Cost-Complexity Pruning):** Grows the full tree first, then prunes subtrees that fail to lower complexity-penalized misclassification costs ($R_■lpha(T) = R(T) + ■lpha|T|$).

• **Clinical Setting Recommendation: Post-Pruning** is superior. It ensures subtle, complex interactions between borderline biomarkers are fully mapped out before removing insignificant or noisy nodes, preventing critical diagnostic signals from being cut too early.

Task 3: Statistical Learning for Risk Stratification

(a) Statistical Model Comparison (Logistic Regression vs Decision Tree):

Logistic Regression maps a linear combination of features to continuous risk probabilities using the sigmoid function: $P(Y=1|X) = 1 / (1 + e^{-■eta X})$.

• **Performance Contrast:** Logistic Regression typically achieves a higher **AUC-ROC (0.89)** than an unpruned Decision Tree (0.82). The decision tree generates rigid step-like boundaries, whereas Logistic Regression produces smooth, continuous risk curves.

- **Preferability:** Statistical models are preferred when the clinical objective requires well-calibrated, probabilistic risk screening tools (e.g., '34% risk' vs a rigid 'yes/no' label) or when the primary clinical drivers relate linearly with risk factors.

(b) Feature Importance & Cross-Model Agreement:

- **Decision Tree Top Predictors:** 1. Glucose, 2. BMI, 3. Age (determined by total split impurity reduction).
- **Logistic Regression Top Predictors:** 1. Glucose, 2. BMI, 3. Family History (determined by the absolute magnitude of standardized Wald z-scores).

• **Synthesis & Agreement:** Both models agree on Glucose and BMI as the primary risk drivers. They diverge on the third predictor due to structural differences: the Decision Tree captures non-linear age thresholds, while Logistic Regression captures the linear risk added by a positive family history.

Task 4: Reinforcement Learning for Treatment Recommendation

(a) MDP Formulation:

- **States (S):** Discrete patient metabolic tiers determined by clinical metrics: S_0 (Normal), S_1 (Pre-diabetic), S_2 (Uncontrolled Diabetes).
- **Actions (A):** Clinical therapeutic interventions: A_0 (Diet), A_1 (Exercise), A_2 (Medication), A_3 (Continuous Monitoring).
- **Reward Function (R):** Numeric feedback reflecting patient health changes: +10 for transitioning to a safer state (e.g., $S_1 ightarrow S_0$), -20 for worsening metabolic health ($S_1 ightarrow S_2$), and a penalty (-5) for high medication doses if lifestyle changes suffice.
- **Transition Dynamics ($P(S'|S,A)$):** Probabilistic model encoding how a patient responds to treatments over time. For example, a patient in state S_1 taking action A_0 (Diet) has a 60% probability of moving to S_0 and a 40% probability of remaining in S_1.

(b) Q-Learning Mathematical Execution:

Updates leverage the temporal difference Bellman optimality equation: $Q(s,a) \leftarrow Q(s,a) + \alpha [R(s,a) + \gamma \max_{a'} Q(s', a') - Q(s,a)]$. Let learning rate $\alpha = 0.2$ and discount factor $\gamma = 0.9$. Assuming all values start at 0:

Iteration	State (s)	Action (a)	Reward (R)	Next State (s')	Updated $Q(s,a)$ Formula & Value
1	S1 (Pre-diabetic)	A0 (Diet)	+10	S0	0 + 0.2*[10 + 0.9*0 - 0] = 2.00
1	S2 (Uncontrolled)	A2 (Med)	+5	S1	0 + 0.2*[5 + 0.9*0 - 0] = 1.00
2	S2 (Uncontrolled)	A2 (Med)	+5	S1	1.0 + 0.2*[5 + 0.9*2.0 - 1.0] = 2.16

Convergence Criterion: The Q-table is considered fully trained when the maximum absolute change across all state-action values falls below a small threshold ($\max_{s,a} |Q_{t+1}(s,a) - Q_t(s,a)| < \epsilon$, where $\epsilon = 10^{-4}$) over consecutive training epochs.

(c) Policy Extraction & Ethical Considerations:

The optimal policy is extracted by selecting the action that maximizes the learned utility for each state: $\pi^*(s) = \arg\max_a Q(s,a)$.

- **Patient Safety & Risk:** Exploration steps (like epsilon-greedy actions) can lead to unsafe recommendations if the agent tries sub-optimal actions on a real patient. Training must be restricted to highly accurate simulators or offline clinical data.
- **Algorithmic Bias:** If the historical training data lacks representation for specific demographic groups, the reward functions and policy recommendations could become biased and deliver lower quality care to minority populations.
- **Explainability:** Deep reinforcement learning models or dense Q-tables do not inherently explain *why* an intervention sequence is optimal, making clinical audit and physician validation challenging.